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VTU Question Papers

“Electronics & Communication Engineering”

III, V & VII Semester

Dec/Jan - 2024

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**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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1	BEC302	Digital System Design Using Verilog	Dec 23/Jan 24
2	BEC303	Electronic Principles and Circuits	Dec 23/Jan 24
3	BEC304	Network Analysis	Dec 23/Jan 24
4	BMATEC301/BEC/ BBM301	AV Mathematics – III for EC/BM Engineering	Dec 23/Jan 24
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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Digital System Design Using Verilog

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Develop a truth table of logic which takes two , 2 – bit binary numbers as its input and generate on output equal to 1, when the sum of the two numbers is odd.	7	L1	CO1
	b.	Convert the following Boolean function into : i) $f(abc) = (\bar{a} + b)(b + \bar{c})$ – Min term canonical form. ii) $f(xyz) = x + \bar{x}\bar{z}(y + \bar{z})$ – Max term canonical form.	7	L1	CO1
	c.	List the difference between Prime implicant and Essential prime implicant.	6	L1	CO1
OR					
Q.2	a.	Simplify the given Boolean function using Quine Mc Cluskey Minimization Technique for the function $R = f(abcd) = \Sigma(0, 1, 2, 6, 7, 9, 10, 12) + dc(3, 5)$.	10	L1	CO1
	b.	Find the minimal sum and minimal product for the given function using K – map method for the function $R = f(abcd) = \Sigma m(0, 1, 3, 7, 8, 12) + dc(5, 10, 13, 14)$.	10	L1	CO1
Module – 2					
Q.3	a.	List the difference between decoder and encoder and implement full adder using IC – 74138.	10	L3	CO2
	b.	What is Comparator? Design a 2 – bit digital comparator.	10	L3	CO2
OR					
Q.4	a.	Realize the Boolean function $P = f(wxyz) = \Sigma(0, 1, 5, 6, 7, 10, 15)$ using i) 8 : 1 MUX ii) 4 : 1 MUX.	10	L2	CO2
	b.	With neat logic diagram, explain carry ahead adder.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the working of master slave JK flip flop with help of Logic diagram , Function table , Logic symbol and Timing diagram.	10	L1	CO3
	b.	Obtain the characteristic equation for : i) SR flip - flop ii) J – K – flip - flop iii) D – flip - flop iv) T – flip - flop.	10	L2	CO3
OR					

Q.6	a.	Design a Synchronous 3 – bit up counter using J K – flip - flop.	10	L4	CO3															
	b.	Design a 4 – bit universal shift register using positive edge triggered D – flip - flop and 4 : 1 MUX , to operate as shown in table below : <table border="1"><tr><td>S₁</td><td>S₀</td><td>Register Operation</td></tr><tr><td>0</td><td>0</td><td>- Hold</td></tr><tr><td>0</td><td>1</td><td>- Shift right</td></tr><tr><td>1</td><td>0</td><td>- Shift left</td></tr><tr><td>1</td><td>1</td><td>- Parallel load operation</td></tr></table>	S ₁	S ₀	Register Operation	0	0	- Hold	0	1	- Shift right	1	0	- Shift left	1	1	- Parallel load operation	10	L4	CO3
S ₁	S ₀	Register Operation																		
0	0	- Hold																		
0	1	- Shift right																		
1	0	- Shift left																		
1	1	- Parallel load operation																		
Module – 4																				
Q.7	a.	Illustrate the structure and verilog module and write a verilog code for Half – adder using structural model.	10	L3	CO4															
	b.	What are different types of operators used in HDL with example?	10	L2	CO4															
OR																				
Q.8	a.	Illustrate the structure of Data flow description with example.	10	L3	CO4															
	b.	Write the syntax of conditional signal assignment statement. Write a code for 4 : 1 MUX using conditional signal statement.	10	L2	CO4															
Module – 5																				
Q.9	a.	Write the structure of Verilog behavioral description.	6	L2	CO4															
	b.	Write the syntax of IF statement with example.	7	L2	CO4															
	c.	Write a code for D – Latch using Behavioral description.	7	L2	CO4															
OR																				
Q.10	a.	Write the syntax of While loop statement with example.	10	L2	CO4															
	b.	Write a verilog code of a 3 – bit ripple carry adder using Structural description method.	10	L2	CO4															

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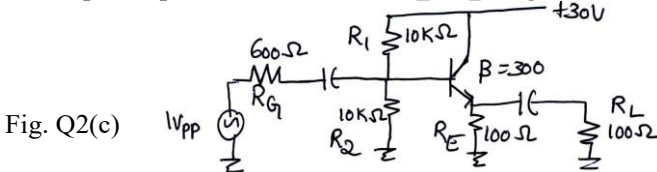
Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Electronic Principles and Circuits

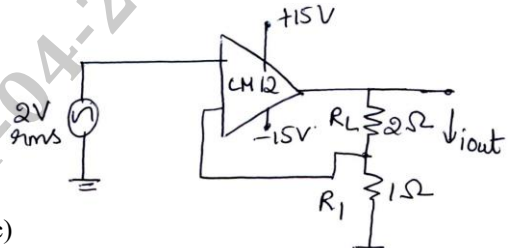
Time: 3 hrs.

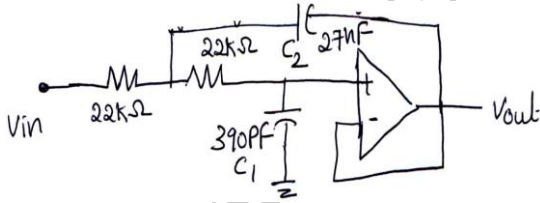
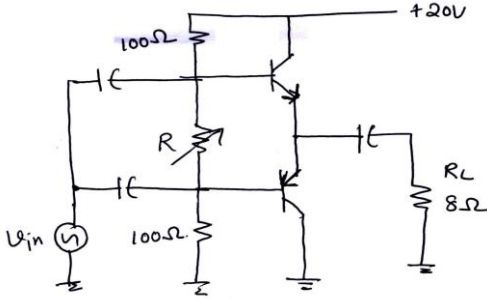
Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the simplified analysis of a voltage divider bias circuit of a transistor. Also list the steps in analysis.	8	L1	CO1
	b.	Analyze a VDB Amplifier circuit with respect to DC circuit, AC - π model, AC – T model.	7	L3	CO1
	c.	Design a positive and negative biased clipper circuit.	5	L3	CO1
OR					
Q.2	a.	With the importance of Coupling capacitor, explain the Base – Biased amplifier circuit. Support your answer with base current , collector current and collector voltage. Also draw its voltage waveforms.	10	L3	CO1
	b.	Explain the basic idea of Common – Collector (CC) amplifier. Give the mathematical relation of AC. Emitter resistance (r_e) , Voltage Gain (A_v) , Input impedance of the base ($Z_{in(base)}$) and Input impedance of the stage ($Z_{in(stage)}$).	6	L2	CO1
	c.	Calculate the output impedance for the circuit below, given $V_{BQ} = 15V$. 	4	L2	CO1
Module – 2					
Q.3	a.	Biasing by fixing V_{GS} is not a good approach to bias a MOSFET. Why? Explain biasing by fixing V_G and connecting a resistance in the source for MOSFET.	8	L2	CO2
	b.	Design a fixed V_G and resistance in the source biasing circuit, to establish drain current $I_D = 0.5mA$, $V_t = 1V$, $K_n^1 W/K = 1mA/V^2$, $\lambda = 0$. Use power supply $V_{DD} = 15V$.	5	L3	CO2
	c.	Obtain the transfer and drain characteristics of n – channel MOSFET and calculate Drain resistance (r_d) , Mutual conductance (g_m) and Amplification factor (μ).	7	L2	CO2
OR					
Q.4	a.	Illustrate the development of T – equivalent circuit model for the MOSFET.	6	L2	CO2

	b.	Draw and explain the small signal equivalent model for Common – Source amplifier without source resistance and write the equation for R_{in} , R_{out} , A_v and G_v .	8	L2	CO2
	c.	For a Common Gate (CG) amplifier circuit, given $g_m = 1\text{mA/V}$, $R_D = 15\text{k}\Omega$, $R_L = 15\text{k}\Omega$, $R_{sig} = 50\Omega$, $R_G = 4.7\mu\Omega$. Find R_{in} , R_{out} , A_{vo} , A_v and G_v .	6	L2	CO2
Module – 3					
Q.5	a.	Explain how an Op – amp summer circuit be configured to function as a subtractor.	5	L1	CO3
	b.	How does the design and configuration of an Op – amp R/2R DAC contribute to its accuracy and performance in converting digital signals to analog signals?	8	L2	CO3
	c.	Design and draw the frequency response of common source JFET / MOSFET amplifier.	7	L2	CO3
OR					
Q.6	a.	Describe the working of inverting Schmitt trigger circuit. How is Schmitt trigger different from regular comparator circuit? Explain with the help of Hysteresis curve.	8	L2	CO3
	b.	Explain the working of Colpitts Oscillator with CE connection.	6	L2	CO3
	c.	Explain the Monostable operation of 555 timers.	6	L2	CO3
Module – 4					
Q.7	a.	Explain the four types of Negative feedback amplifier.	8	L1	CO4
	b.	Explain the VCVS amplifier. Obtain its exact closed – loop voltage gain and Ideal Closed – Loop Voltage gain. Also define Gain stability, Closed loop input impedance and Closed loop output impedance of a VCVS amplifier.	8	L2	CO4
	c.	Calculate the load power, load current for the given VCIS amplifier circuit.	4	L2	CO4
		 Fig. Q7(c)			
OR					
Q.8	a.	Explain the Ideal response of filters.	8	L1	CO4

	b.	Determine the pole frequency , Q , Cutoff frequency and 3 – dB frequency, for the filter circuit given below :	5	L2	CO4
		Fig. Q8(b)  Given $K_0 = 0.99$, $K_C = 1.38$, $K_3 = 1.54$.			
	c.	Design a Halfwave and Fullwave precision rectifier using Op – amp.	7	L3	CO4
Module – 5					
Q.9	a.	Explain class A amplifier , interns of its power gain, Output power , Power dissipation and efficiency.	8	L1	CO5
	b.	Explain class B push pull emitter follower amplifier. How can the crossover distortion be eliminated?	8	L1	CO5
	c.	Calculate the maximum transistor power dissipation and maximum output power for the given circuit.	4	L2	CO5
		Fig. Q9(c) 			
OR					
Q.10	a.	What is an SCR? With the help of basic SCR circuit, explain the gate triggering.	6	L1	CO5
	b.	Explain the phase control method of TRIAC, along with the voltage waveforms.	7	L1	CO5
	c.	Design a full wave controlled rectifier circuit using RC triggering.	7	L3	CO5

Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

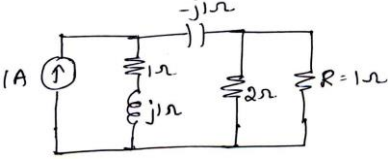
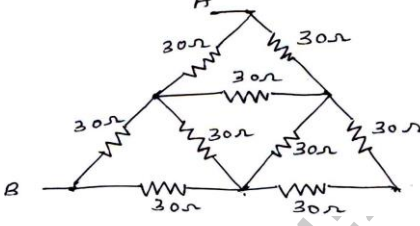
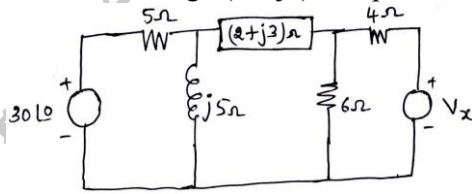
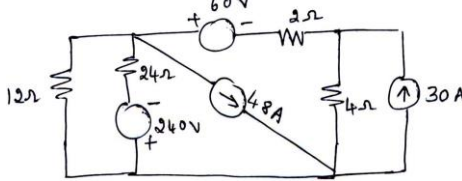
Network Analysis

Time: 3 hrs.

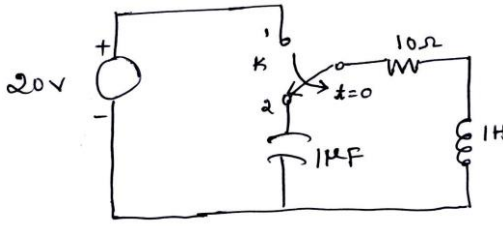
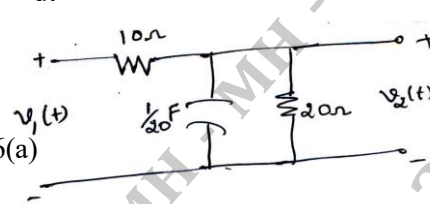
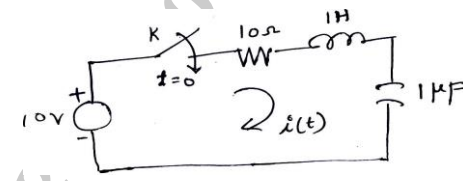
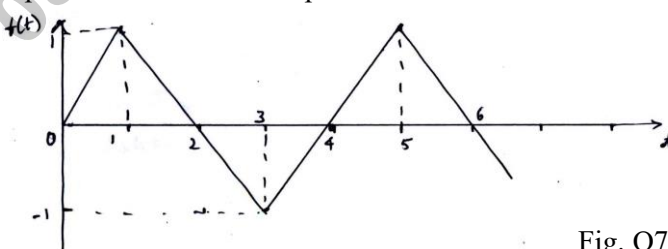
Max. Marks: 100

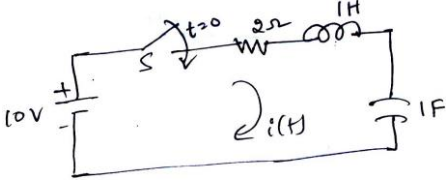
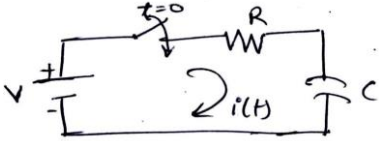
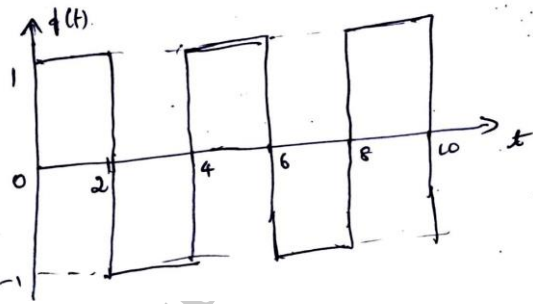
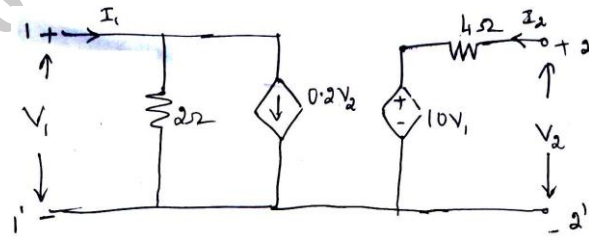
Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the classification of electrical networks.	8	L2	CO1
	b.	For the network shown in Fig. Q1(b), find the current through load resistor 'R' using loop analysis. 	6	L3	CO1
	c.	For the network shown in Fig. Q1(c), find the equivalent resistance between the terminals A – B using Star – Delta transformation. 	6	L3	CO1
OR					
Q.2	a.	Derive an expression for the equivalent impedances between the terminals for Delta – Star transformation.	6	L2	CO1
	b.	Use modal analysis to find the value of voltage 'V _x ' in the circuit shown in Fig. Q2(b), such that the current through (2 + j3)Ω impedance is zero. 	7	L3	CO1
	c.	Determine the current through 12Ω resistor shown in Fig. Q2(c), using Source Shifting / Transformation method. 	7	L3	CO1

Module – 2					
Q.3	a.	Using Superposition theorem, obtain the current 'I' for the network shown in Fig. Q3(a).	10	L2	CO1
		Fig. Q3(a)			
	b.	Using Millman's theorem, calculate the current through the load in the circuit shown in Fig. Q3(b).	10	L3	CO2
		Fig. Q3(b)			
OR					
Q.4	a.	State and explain Norton's theorem.	6	L2	CO2
	b.	For the network shown in Fig. Q4(b), find the current through 16Ω resistor using Thevenin's theorem.	7	L3	CO2
		Fig. Q4(b)			
	c.	For the network shown in Fig. Q4(c), find the value of Z_L for which maximum power transfer occurs. Also find the maximum power.	7	L3	CO2
		Fig. Q4(c)			
Module – 3					
Q.5	a.	Explain the initial and final conditions in basic elements.	6	L2	CO3

	b. For the circuit shown in Fig. Q5(b), the switch 'K' is changing the position from 1 to 2 at $t = 0$. Steady state condition has been reached at position 1. Find the value of i, $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t = 0^+$.  Fig. Q5(b)	8	L3	CO3
	c. Obtain an expression for transient response $i(t)$ of a series R – L circuit when excited by DC supply.	6	L2	CO3
OR				
Q.6	a. In the circuit shown in Fig. Q6(a), $v_1(t) = e^{-t}$ for $t \geq 0$ and zero for all $t < 0$. If the capacitor is initially uncharged, determine the value of $v_2(t)$, $\frac{dv_2(t)}{dt}$, $\frac{d^2v_2(t)}{dt^2}$ and $\frac{d^3v_2(t)}{dt^3}$ at $t = 0^+$.  Fig. Q6(a)	10	L3	CO3
	b. For the circuit shown in Fig. Q6(b), the switch is closed at $t = 0$. Determine i, $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ and $\frac{d^3i}{dt^3}$ at $t = 0^+$.  Fig. Q6(b)	10	L3	CO3
Module – 4				
Q.7	a. State and prove Initial Value Theorem.	6	L2	CO3
	b. Find the Laplace Transform of the periodic waveform shown in Fig. Q7(b).  Fig. Q7(b)	8	L3	CO3

	c.	Using Laplace transform, determine the current $i(t)$ in the circuit shown in Fig. Q7(c), when the switch 'S' is closed at $t = 0$. Assume zero initial conditions.	6	L3	CO3
		Fig. Q7(c) 			
OR					
Q.8	a.	State and prove differentiate by 'S' domain property.	6	L2	CO3
	b.	In the circuit shown in Fig. Q8(b), the switch is closed at $t = 0$. Obtain the expression for the current.	6	L3	CO3
		Fig. Q8(b) 			
	c.	Obtain the Laplace Transform of the square wave shown in Fig. Q8(c).	8	L3	CO3
		Fig. Q8(c) 			
Module – 5					
Q.9	a.	What are Impedance and Hybrid parameters? Derive the expression for the same.	8	L2	CO4
	b.	Derive an expression for Transmission parameters in terms of Z – parameters.	5	L2	CO4
	c.	For the circuit shown in Fig. Q9(c), find Y – parameters.	7	L3	CO4
		Fig. Q9(c) 			
OR					
Q.10	a.	Derive an expression for bandwidth of a series Resonant circuit.	7	L2	CO5

	b. A series RLC circuit consists of a resistance of $1\text{ k}\Omega$ and an inductance of 100mH in series with capacitance of 10PF connected across 100V supply. Determine i) Resonant frequency ii) Quality factor iii) Maximum current in the circuit iv) Bandwidth v) Half power frequencies v) Selectivity factor.	7	L3	CO5
	c. For the circuit shown in Fig. Q10(c), find i) Resonant frequency ii) Quality factor iii) Bandwidth iv) Impedance at resonance v) Current at resonance.	6	L3	CO5

Fig. Q10(c)

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BMATEC301/BEC/BBM301

Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 AV Mathematics – III for EC/BM Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. Statistical table and handbook permitted.
4. Use of VTU Mathematics handbook is permitted.*

Module – 1				M	L	C														
Q.1	a.	Obtain the Fourier series of $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2\pi$. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.	6	L2	CO1															
	b.	Expand $f(x) = 2x-1$ as a Cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1															
	c.	Compute the First harmonics of the Fourier series of $f(x)$. Given the table <table><tr><td>x</td><td>0</td><td>$\frac{\pi}{3}$</td><td>$\frac{2\pi}{3}$</td><td>π</td><td>$\frac{4\pi}{3}$</td><td>$\frac{5\pi}{3}$</td><td>2π</td></tr><tr><td>f(x)</td><td>1.0</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1.0</td></tr></table>	x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π	f(x)	1.0	1.4	1.9	1.7	1.5	1.2	1.0	7	L3
x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π													
f(x)	1.0	1.4	1.9	1.7	1.5	1.2	1.0													
OR																				
Q.2	a.	Obtain the Fourier series of $f(x) = x $ in $(-\ell, \ell)$.	6	L2	CO1															
	b.	Obtain the Cosine half range Fourier series of $f(x) = x^2$ in $0 < x < \pi$.	7	L3	CO1															
	c.	Express Y as a Fourier Cosine series upto second harmonics. Given the table : <table><tr><td>x</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>y</td><td>9.0</td><td>18.2</td><td>24.4</td><td>27.8</td><td>27.5</td><td>22.0</td><td>9.0</td></tr></table>	x	0	2	4	6	8	10	12	y	9.0	18.2	24.4	27.8	27.5	22.0	9.0	7	L3
x	0	2	4	6	8	10	12													
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0													
Module – 2																				
Q.3	a.	Find the Fourier transform of $f(x) = e^{- x }$.	6	L2	CO2															
	b.	Find the Fourier Cosine and Sine transform of $f(x) = e^{-\alpha x}$, $\alpha > 0$.	7	L3	CO2															
	c.	i) Find a Discrete Fourier transform of the single $f = [3, 4, 5, 5]^T$. ii) Find the Inverse Discrete Fourier transform of the single obtained in part (i).	7	L3	CO2															
OR																				
Q.4	a.	Find the Fourier transform of $f(x) = \begin{cases} 1 - x & \text{for } x \leq 1 \\ 0 & \text{for } x > 1 \end{cases}$ and hence deduce that $\int_0^\infty \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$.	6	L2	CO2															

	b.	Obtain the Fourier Cosine transform of $f(x) = \begin{cases} 4x & , 0 < x < 1 \\ 4-x & , 1 < x < 4 \\ 0 & , x > 4 \end{cases}$	7	L3	CO2
	c.	Solve the Integral equation $\int_0^\infty f(\theta) \cos \alpha \theta d\theta = \begin{cases} 1-\alpha & , 0 \leq \alpha \leq 1 \\ 0 & , \alpha > 1 \end{cases}$ and hence evaluate $\int_0^\infty \frac{\sin^2 t}{t^2} dt$.	7	L3	CO2
Module – 3					
Q.5	a.	Find the Z – transform of i) $\cos n\theta$ ii) $\sin n\theta$.	6	L2	CO3
	b.	Find the Inverse Z – transform of $\frac{z^2 - 8z}{(z-4)^2}$.	7	L3	CO3
	c.	Solve the difference equation $y_{n+2} - 4y_n = 0$. Given that $y_0 = 0$ and $y_1 = 2$.	7	L3	CO3
OR					
Q.6	a.	Find the Z – transform of $2n + \sin\left(\frac{n\pi}{4}\right) + 1$.	6	L2	CO3
	b.	Compute the Inverse Z – transform of $\frac{3z^2 + 2z}{(5z-1)(5z+2)}$.	7	L3	CO3
	c.	Solve the difference equation $u_{n+2} + 6u_{n+1} + 9u_n = 2^n$ with $u_0 = u_1 = 0$, using Z – transforms.	7	L3	CO3
Module – 4					
Q.7	a.	Solve $(D^4 - m^4)y = 0$.	6	L2	CO4
	b.	Solve $(D^2 - 2D + 1)y = \sin x + e^x$.	7	L3	CO4
	c.	Solve $x \frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} = \frac{1}{x}$.	7	L3	CO4
OR					
Q.8	a.	Solve $\frac{d^3y}{dx^3} + 8y = x^4 + 2x + 1$.	6	L2	CO4
	b.	Solve the Legendre's form of Linear equation. $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin 2 [\log(1+x)]$.	7	L3	CO4
	c.	In the LCR circuit the charge q on a plate of condenser is given by $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E \sin p t$. Solve the above equation.	7	L3	CO4

Module – 5																																						
Q.9	a.	Find a Least square straight line for the following data : <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>6</td><td>4</td><td>3</td><td>5</td><td>4</td><td>2</td></tr></table>	x	1	2	3	4	5	6	y	6	4	3	5	4	2	6	L2	CO5																			
x	1	2	3	4	5	6																																
y	6	4	3	5	4	2																																
	b.	In a partially destroyed laboratory record, the lines of regression of y on x and x on y are available as $4x - 5y + 33 = 0$ and $20x - 9y = 107$. Calculate $\bar{x}$, $\bar{y}$ and coefficient of correlation between x and y.	7	L3	CO5																																	
	c.	Ten competition in a beauty contest are ranked by two judges A and B in the following order. Calculate the rank correlation coefficient. <table><tr><td>ID No. of competition</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Judge A</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>Judge B</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table>	ID No. of competition	1	2	3	4	5	6	7	8	9	10	Judge A	1	6	5	10	3	2	4	9	7	8	Judge B	6	4	9	8	1	2	3	10	5	7	7	L3	CO5
ID No. of competition	1	2	3	4	5	6	7	8	9	10																												
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OR																																						
Q.10	a.	Fit a parabola for the data in the form $y = ax^2 + bx + c$. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>10</td><td>12</td><td>13</td><td>16</td><td>19</td></tr></table>	x	1	2	3	4	5	y	10	12	13	16	19	6	L2	CO5																					
x	1	2	3	4	5																																	
y	10	12	13	16	19																																	
	b.	The following table gives the heights of Father (x) and Sons (y) : <table><tr><td>x</td><td>65</td><td>66</td><td>67</td><td>67</td><td>68</td><td>69</td><td>70</td><td>72</td></tr><tr><td>y</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table> Find the lines of regression and hence calculate the co-efficient of correlation.	x	65	66	67	67	68	69	70	72	y	67	68	65	68	72	72	69	71	7	L3	CO5															
x	65	66	67	67	68	69	70	72																														
y	67	68	65	68	72	72	69	71																														
	c.	Determine the rank correlation for the following data which shows the marks obtained in two quizzes in mathematics. <table><tr><td>Marks in first quiz X</td><td>6</td><td>5</td><td>8</td><td>8</td><td>7</td><td>6</td><td>10</td><td>4</td><td>9</td><td>7</td></tr><tr><td>Marks in first quiz Y</td><td>8</td><td>7</td><td>7</td><td>10</td><td>5</td><td>8</td><td>10</td><td>6</td><td>8</td><td>6</td></tr></table>	Marks in first quiz X	6	5	8	8	7	6	10	4	9	7	Marks in first quiz Y	8	7	7	10	5	8	10	6	8	6	7	L3	CO5											
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Marks in first quiz Y	8	7	7	10	5	8	10	6	8	6																												

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024 Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
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4. Darkening two circles for the same question makes the answer invalid.
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-
1. Biodiversity is a measure of variation at the _____ level
 a) Genetic b) Species c) Ecosystem d) All of these
 2. World Environment Day is celebrated on _____
 a) 5th May b) 5th June c) 18th June d) 16th August
 3. Mining means
 a) To conserve minerals
 b) To check pollution
 c) To extract minerals and ores
 d) None of the above
 4. Direct conversion of solar energy is attained by
 a) Solar Photo volcanic system
 b) Solar diesel hybrid system
 c) Solar thermal system
 d) Solar air heater
 5. What % of its geographical area of a country should be under forest cover
 a) 23% b) 43% c) 13% d) 33%
 6. Hazardous Waste Management Act was enacted in India in the year
 a) 1988 b) 1989 c) 1990 d) 1991
 7. Which of these following elements is the case of e-waste?
 a) Cadmium b) Beryllium c) Lead d) All of these

8. Remote sensing techniques make use of the properties of _____ emitted, reflected or diffracted by the sensed objects
 - a) Electric waves
 - b) Sound waves
 - c) Electromagnetic waves
 - d) Wind waves
9. The altitudinal distance of a geostationary satellite from the earth is about
 - a) 26,000km
 - b) 30,000km
 - c) 36,000km
 - d) 44,000km
10. Montreal protocol is related to the
 - a) Food security
 - b) Global warming
 - c) Sustainable development
 - d) Ozone layer depletion
11. The main cause of damage to Taj Mahal is _____
 - a) Water pollution
 - b) Soil pollution
 - c) Acid rain
 - d) Fog
12. Reducing the amount of future climate change is called.
 - a) Mitigation
 - b) Geo-engineering
 - c) Adaptation
 - d) None of these
13. Ozone layer is at a height of _____ above the Earth's surface
 - a) 19 to 48m
 - b) 19 to 480m
 - c) 19 to 48km
 - d) 190 to 480km
14. Which ministry is mainly responsible for research and development in renewable energy sources such as wind, power small hydro, biogas and solar power
 - a) Human Resource Development
 - b) Agriculture and Famous welfare
 - c) Ministry of new and Renewable energy
 - d) Health and Family welfare
15. The OTEC is an energy technology that converts
 - a) Energy in large fides of ocean to generate electricity
 - b) Energy in ocean waves to generate electricity
 - c) Energy in ocean due to thermal gradient to generate electricity
 - d) Energy in the fast moving ocean currents to generate electricity
16. In a Lake, phytoplankton grow in abundance in
 - a) Littoral zone
 - b) Limnetic zone
 - c) Profundal zone
 - d) Benthic region
17. The prescribed limits of noise in residential area during day is
 - a) 55dB
 - b) 45dB
 - c) 60dB
 - d) 50dB
18. The maximum allowable concentration of fluorides in drinking water
 - a) 3mg/L
 - b) 2mg/L
 - c) 2.5mg/L
 - d) 1.5mg/L
19. The color code of plastic bag for disposing microbial laboratory culture waste
 - a) Red
 - b) Black
 - c) Blue
 - d) White
20. The hazardous pollutant released from batteries is
 - a) Arsenic
 - b) Cobalt
 - c) Barium
 - d) Cadmium
21. GIS uses the information from which of the following sources
 - a) Non-Spatial Information System
 - b) Spatial Information System
 - c) Global Information System
 - d) Position Information System

22. EIA can be expanded as
 a) Environment and Industrial Act
 b) Environmental and Impact Activities
 c) Environmental Impact Assessment
 d) Environmental Impact Activity
23. ISO 14000 standards deals with
 a) Pollution management
 b) Risk management
 c) Environmental management
 d) None of these
24. Which of the following represents India in ISO
 a) PFRDA
 b) FSSAI
 c) BIS
 d) BCCI
25. Which of the following is having high population density
 a) India
 b) China
 c) USA
 d) Western Europe
26. Environment education is targeted to
 a) General public
 b) Professional social groups
 c) Technical and Scientists
 d) All of the above
27. Discharge of municipal waste cause
 a) Depletion of dissolved oxygen
 b) Destroy aquatic life
 c) Impair biological activity
 d) All of the above
28. _____ is are referred to a Earth's lungs
 a) Forests
 b) Carbon cycle
 c) Water sources
 d) Miner
29. Solid waste is best managed through
 a) Incineration
 b) Open dumping
 c) Sanitary landfill
 d) Composting
30. Love canal tragedy is attributed to
 a) Soil pollution
 b) Hazardous waste
 c) Air pollution
 d) None of these
31. According to Biomedical Waste (Management and Handling) Rules 1998, waste should not be stored beyond _____
 a) 12 hours
 b) 48 hours
 c) 72 hours
 d) 96 hours
32. Pyrolysis is an _____ process
 a) Exothermic
 b) Endothermic
 c) Both a and b
 d) Neither a and b
33. Chlorofluorocarbons are
 a) Nontoxic
 b) Flammable
 c) Corrosive
 d) Odorous
34. Which of the following is an air pollutant
 a) Carbon dioxide
 b) Oxygen
 c) Nitrogen
 d) Particulate matter
35. Urbanization is
 a) Local environmental issue
 b) Nation environmental issue
 c) Both a and b
 d) Not at all an issue

36. Earth day is held every year on :
 a) June 5th b) November 23rd c) April 22nd d) January 26th
37. The term hotspot was introduced by –
 a) Norman Myere b) Jacob Von Verkul
 c) A.G. Transley d) Ernst Haeckel
38. In an Ecosyste, the energy flow is always
 a) Always unidirectional b) Always bidirectional
 c) In any direction d) Always down directional
39. Which of the following is considered as an alternate fuel
 a) CNG b) Kerosene c) Coal d) Petrol
40. Nuclear power plant in Karnataka is located at
 a) Bhadravati b) Sandur c) Raichur d) Kaiga
41. Disaster management includes _____
 a) Mitigation b) Reconstruction c) Rehabilitation d) All of these
42. What is the health effects of fluoride in drinking waster
 a) Arthritis b) Diarrhea c) Anemia d) All of these
43. What is the permissible range of pH for drinking water as per Indian standards
 a) 6 – 9 b) 6 – 8.5 c) 6.5 – 8.5 d) 6 5 – 7.5
44. The infiltration of water into the subsurface is the
 a) Influent b) Effluent c) Discharge d) Recharge
45. Environmental (Protection Act) was enacted in the year
 a) 1986 b) 1992 c) 1984 d) 1974
46. What is the full form of NGO
 a) Non-Governmental Organization
 b) Non-Governance Organizations
 c) No- Governance Organizations
 d) Null – Governmental organizations
47. The primary cause of acid rain around the world is
 a) CFC b) SO₂ c) CO d) O₃
48. Bhopal Gas Tragedy caused due to leakage of
 a) Methyl ISO Cyanate (MIC) b) Sulphur dioxide
 c) Mustered gas d) Methane
49. Deforestation can
 a) Increase the rainfall b) Increase soil fertility
 c) Introduce silt in the river d) None of these
50. The word Environment is derived from
 a) Greek b) French c) Spanish d) English

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan. 2024
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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-
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- 50 Love canal tragedy is attributed to –
 a) Soil pollution b) Hazardous waste c) Air pollution d) None of these

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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan. 2024
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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- 49 The color code of plastic bag for disposing microbial laboratory culture waste
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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan. 2024
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
- 1 GIS uses the information from which of the following sources
 - a) Non-Spatial Information System
 - b) Spatial Information System
 - c) Global Information System
 - d) Position Information System

 - 2 EIA can be expanded as
 - a) Environment and Industrial Act
 - b) Environmental and Impact Activities
 - c) Environmental Impact Assessment
 - d) Environmental Impact Activity

 - 3 ISO 14000 standards deals with

a) Pollution management	b) Risk management
c) Environmental management	d) None of these

 - 4 Which of the following represents India in ISO

a) PFRDA	b) FSSAI	c) BIS	d) BCCI
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 - 5 Which of the following is having high population density

a) India	b) China	c) USA	d) Western Europe
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 - 6 Environment education is targeted to
 - a) General public
 - b) Professional social groups
 - c) Technical and Scientists
 - d) All of the above

- 7 Discharge of municipal waste cause
 a) Depletion of dissolved oxygen
 b) Destroy aquatic life
 c) Impair biological activity
 d) All of the above
- 8 _____ is are referred to a Earth's lungs
 a) Forests b) Carbon cycle c) Water sources d) Miner
- 9 Solid waste is best managed through
 a) Incineration b) Open dumping c) Sanitary landfill d) Composting
- 10 Love canal tragedy is attributed to
 a) Soil pollution b) Hazardous waste c) Air pollution d) None of these
- 11 Disaster management includes _____
 a) Mitigation b) Reconstruction c) Rehabilitation d) All of these
- 12 What is the health effects of fluoride in drinking waster
 a) Arthritis b) Diarrhea c) Anemia d) All of these
- 13 What is the permissible range of pH for drinking water as per Indian standards
 a) 6 – 9 b) 6 – 8.5 c) 6.5 – 8.5 d) 6.5 – 7.5
- 14 The infiltration of water into the subsurface is the
 a) Influent b) Effluent c) Discharge d) Recharge
- 15 Environmental (Protection Act) was enacted in the year
 a) 1986 b) 1992 c) 1984 d) 1974
- 16 What is the full form of NGO
 a) Non-Governmental Organization
 b) Non-Governance Organizations
 c) No- Governance Organizations
 d) Null – Governmental organizations
- 17 The primary cause of acid rain around the world is
 a) CFC b) SO₂ c) CO d) O₃
- 18 Bhopal Gas Tragedy caused due to leakage of
 a) Methyl ISO Cyanate (MIC) b) Sulphur dioxide
 c) Mustered gas d) Methane
- 19 Deforestation can
 a) Increase the rainfall
 b) Increase soil fertility
 c) Introduce silt in the river
 d) None of the above
- 20 The word Environment is derived from
 a) Greek b) French c) Spanish d) English
- 21 According to Biomedical Waste (Management and Handling) Rules 1998, waste should not be stored beyond _____
 a) 12 hours b) 48 hours c) 72 hours d) 96 hours

- 22 Pyrolysis is an _____ process
 a) Exothermic b) Endothermic c) Both a and b d) Neither a and b
- 23 Chlorofluorocarbons are
 a) Nontoxic b) Flammable c) Corrosive d) Odorous
- 24 Which of the following is an air pollutant
 a) Carbon dioxide b) Oxygen c) Nitrogen d) Particulate matter
- 25 Urbanization is
 a) Local environmental issue
 b) Nation environmental issue
 c) Both a and b
 d) Not at all an issue
- 26 Earth day is held every year on :
 a) June 5th b) November 23rd c) April 22nd d) January 26th
- 27 The term hotspot was introduced by –
 a) Norman Myere b) Jacob Von Verkul
 c) A.G. Transley d) Ernst Haeckel
- 28 In an Ecosyste, the energy flow is always
 a) Always unidirectional
 b) Always bidirectional
 c) In any direction
 d) Always down directional
- 29 Which of the following is considered as an alternate fuel
 a) CNG b) Kerosene c) Coal d) Petrol
- 30 Nuclear power plant in Karnataka is located at
 a) Bhadravati b) Sandur c) Raichur d) Kaiga
- 31 The main cause of damage to Taj Mahal is _____
 a) Water pollution b) Soil pollution c) Acid rain d) Fog
- 32 Reducing the amount of future climate change is called.
 a) Mitigation b) Geo-engineering c) Adaptation d) None of these
- 33 Ozone layer is at a height of _____ above the Earth's surface
 a) 19 to 48m b) 19 to 480m c) 19 to 48km d) 190 to 480km
- 34 Which ministry is mainly responsible for research and development in renewable energy sources such as wind, power small hydro, biogas and solar power
 a) Human Resource Development
 b) Agriculture and Famous welfare
 c) Ministry of new and Renewable energy
 d) Health and Family welfare
- 35 The OTEC is an energy technology that converts
 a) Energy in large fides of ocean to generate electricity
 b) Energy in ocean waves to generate electricity
 c) Energy in ocean due to thermal gradient to generate electricity
 d) Energy in the fast moving ocean currents to generate electricity

- 36 In a Lake, phytoplankton grow in abundance in
a) Littoral zone b) Limnetic zone c) Profundal zone d) Benthic region
- 37 The prescribed limits of noise in residential area during day is
a) 55dB b) 45dB c) 60dB d) 50dB
- 38 The maximum allowable concentration of fluorides in drinking water
a) 3mg/L b) 2mg/L c) 2.5mg/L d) 1.5mg/L
- 39 The color code of plastic bag for disposing microbial laboratory culture waste
a) Red b) Black c) Blue d) White
- 40 The hazardous pollutant released from batteries is
a) Arsenic b) Cobalt c) Barium d) Cadmium
- 41 Biodiversity is a measure of variation at the _____ level
a) Genetic b) Species c) Ecosystem d) All of these
- 42 World Environment Day is celebrated on _____
a) 5th May b) 5th June c) 18th June d) 16th August
- 43 Mining means
a) To conserve minerals b) To check pollution
c) To extract minerals and ores d) None of these
- 44 Direct conversion of solar energy is attained by
a) Solar Photo volcanic system
b) Solar diesel hybrid system
c) Solar thermal system
d) Solar air heater
- 45 What % of its geographical area of a country should be under forest cover
a) 23% b) 43% c) 13% d) 33%
- 46 Hazardous Waste Management Act was enacted in India in the year
a) 1988 b) 1989 c) 1990 d) 1991
- 47 Which of these following elements is the case of e-waste?
a) Cadmium b) Beryllium c) Lead d) All of these
- 48 Remote sensing techniques make use of the properties of _____ emitted, reflected or diffracted by the sensed objects
a) Electric waves b) Sound waves
c) Electromagnetic waves d) Wind waves
- 49 The altitudinal distance of a geostationary satellite from the earth is about
a) 26,000km b) 30,000km c) 36,000km d) 44,000km
- 50 Montreal protocol is related to the
a) Food security b) Global warming
c) Sustainable development d) Ozone layer depletion

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Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Digital Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive the expression for error probability of binary phase shift keying using coherent detection. (08 Marks)
- b. An FSK system transmits binary data at the rate of 2×10^6 bit per sec. During the source of transmission, AWGN of zero mean and two sided power spectral density 10^{-20} W/Hz is added to the signal. The amplitude of received wave for digit 1 or 0 is 1 microvolt. Determine the average probability of symbol error assuming non-coherent detection. (06 Marks)
- c. Explain the concept of M-ary PSK. (06 Marks)

OR

- 2 a. With a neat block diagram, explain non-coherent detection of binary FSK technique. (08 Marks)
- b. Binary data is transmitted over AWGN channel using BPSK at a rate of 1Mbps. It is desired to have average probability of error $p_e \leq 10^{-4}$. Noise PSD = 10^{-12} W/Hz. Determine the average carrier power required at receiver input if the detector is of coherent type. [Assume $\text{erfc}(3.5) = 0.00025$]. (06 Marks)
- c. Explain the generation and detection of DPSK with neat block diagram. (06 Marks)

Module-2

- 3 a. Explain the geometric representation of set of in energy signals as combination of N orthonormal basis function. Illustrate the case of N = 2 and M = 3 with necessary diagrams and expressions. (08 Marks)
- b. Explain the correlation receiver using coherent detection. (06 Marks)
- c. Explain the design of band limited signals with controller ISI-partial response signal. (06 Marks)

OR

- 4 a. Using Gram-Schmidt orthogonalization procedure find the set of orthonormal basis function to represent the signals $s_1(t)$, $s_2(t)$ and $s_3(t)$ as shown in Fig.Q.4(a). Also express each of these signals in terms of set of basis function.

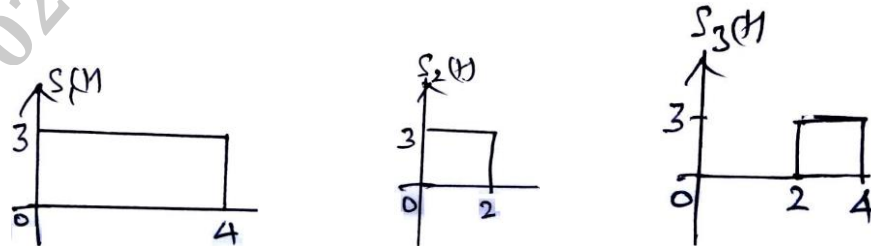


Fig.Q.4(a)

- b. State and prove Nyquist condition for zero ISI.

(10 Marks)

(10 Marks)

Module-3

- 5 a. Explain the model of spread spectrum digital communication system. (10 Marks)
 b. With a neat block diagram, explain the CDMA system band on IS-95. (10 Marks)

OR

- 6 a. Explain the frequency hopped spread spectrum technique with neat transmitter and receiver block diagram. (08 Marks)
 b. The SNR required at the detector to achieve reliable communication in a DSSS communication system is 13dB. If the interference to signal power at the receiver is 20dB. Determine the processing gain required. (04 Marks)
 c. Write a note on application of DS spread spectrum systems. (08 Marks)

Module-4

- 7 a. Define the following with respect to information theory :
 i) Self information
 ii) Entropy
 iii) Source efficiency
 iv) Rate of information. (08 Marks)
 b. Construct binary code for the following source using Shannon's binary encoding procedure.
 $s = \{s_1, s_2, s_3, s_4, s_5\}$ $p = \{0.4, 0.25, 0.15, 0.12, 0.08\}$. (08 Marks)
 c. Explain the types of methods of controlling error. (04 Marks)

OR

- 8 a. Six messages symbols with probability of 0.4, 0.2, 0.2, 0.1, 0.07, 0.03, construct a binary code by using Shannon's Fano encoding procedure. Also determine code efficiency and redundancy. (10 Marks)
 b. A source produces 5 symbols with probabilities of 0.1, 0.3, 0.4, 0.12 and 0.08.
 i) Construct a binary Huffman code
 ii) Determine efficiency and redundancy of the code
 iii) Draw code-tree. (10 Marks)

Module-5

- 9 a. A (7, 4) linear block code having parity matrix $P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$
 i) Find all possible code vector
 ii) Draw the encoding circuit
 iii) Draw the syndrome circuit. (10 Marks)
 b. A (3, 1, 2) convolutional code with $g^{(1)} = (110)$, $g^{(2)} = (101)$ and $g^{(3)} = (111)$.
 i) Draw the encoder block diagram.
 ii) Find the generator matrix.
 iii) Find the code word for information sequence (11101) using transform domain approach. (10 Marks)

OR

- 10 a. For a (2, 1, 4) convolutional encoder as shown in Fig.Q.10(a).

(10 Marks)

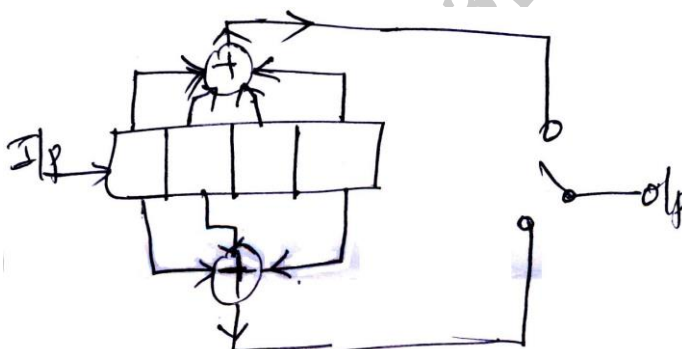


Fig.Q.10(a)

Find the codeword corresponding to the information source (10111). Using time domain and transform domain approach. (10 Marks)

- b. A, (2, 1, 2) binary convolutional encoder as shown in Fig.Q.10(b). Draw the state table, state transition table, state diagram and corresponding code tree, for the message 10111. Find the encoded sequence.

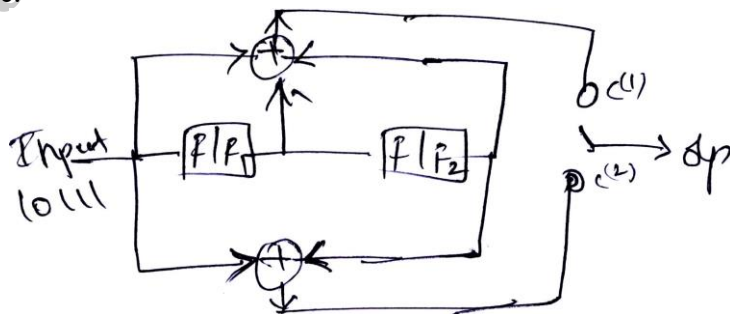


Fig.Q.10(b)

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21EC52

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Computer Organization and Arm Microcontrollers

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, discuss the operational concepts in a computer highlighting the role of PC, MAR, MDR and IR. (10 Marks)
- b. Explain system software functions in computer. (06 Marks)
- c. Explain computer basic performance equation. (04 Marks)

OR

- 2 a. Explain operation of DMA with neat diagram. (10 Marks)
- b. With a neat diagram, discuss implementation of interrupt priority using individual request and acknowledge lines. (06 Marks)
- c. Illustrate with a neat diagram, a computer using different interface standards. (04 Marks)

Module-2

- 3 a. With a neat diagram, explain the internal organization of 16×8 memory chip. (10 Marks)
- b. State and explain the types of read only memory and memory hierarchy. (10 Marks)

OR

- 4 a. With a neat diagram, explain the three bus organization of a datapath. (10 Marks)
- b. Explain basic idea of pipelining and 4-stage pipeline structure. (10 Marks)

Module-3

- 5 a. With a neat diagram, explain the four main hardware components of an ARM based embedded device. (08 Marks)
- b. Discuss ARM design philosophy. (06 Marks)
- c. Explain the factors that make ARM instruction set suitable for embedded applications. (06 Marks)

OR

- 6 a. Explain ARM core data flow model with a neat diagram. (08 Marks)
- b. Explain the different processor modes provided by ARM7. (06 Marks)
- c. Discuss with a neat diagram:
 - i) Von Neumann architecture with cache
 - ii) Harvard architecture with TCM. (06 Marks)

Module-4

- 7 a. Explain with neat diagram, barrel shifter operation in ARM processor. (08 Marks)
- b. Explain with an example the concept of semaphore using swap instruction. (06 Marks)
- c. Develop an assembly language program to multiply two 16-bit numbers. (06 Marks)

OR

- 8 a. Explain the following with example:
i) MSR ii) MVN iii) TST iv) BIC. (08 Marks)
- b. Explain with an example forward and backward branch. (06 Marks)
- c. Develop an assembly language program to find GCD of two numbers using conditional execution. (06 Marks)

Module-5

- 9 a. Discuss with an example code density in thumb instruction set over ARM. (08 Marks)
- b. Explain ARM-thumb interworking. (06 Marks)
- c. Explain with example thumb stack operations. (06 Marks)

OR

- 10 a. Explain with an example the effect of using 'char' and 'short' as local variable types in ARM processor. (08 Marks)
- b. List the C compiler data type mapping for an ARM target with their implementation. (05 Marks)
- c. With an example, compare the efficiencies of signed int and unsigned int with an example. (07 Marks)

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GBCS SCHEME

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21EC53

Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Computer Communication Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Outline the functions of various layers in TCP/IP with necessary diagram to show logical connection between layers. (10 Marks)
- b. Compare various physical topologies in a computer network. (10 Marks)

OR

- 2 a. Explain five components of data communication with a neat diagram. (06 Marks)
- b. Explain different types of data-flow with a neat diagram. (06 Marks)
- c. Explain different types of switched networks used in computer network with relevant diagram. (08 Marks)

Module-2

- 3 a. Explain character-oriented framing and bit-oriented framing with an example. (10 Marks)
- b. With a neat diagram, explain standard Ethernet frame format. (10 Marks)

OR

- 4 a. With a neat flow diagram and timing diagram, explain CSMA/CD. (10 Marks)
- b. A pure ALOHA network transmits 200 bit frames on a shared channel of 200 kbps, what is the throughput if the system produces (all stations together):
 - (i) 1000 frames per second
 - (ii) 500 frames per second
 - (iii) 250 frames per second(06 Marks)
- c. Explain implementation of standard Ethernet. (04 Marks)

Module-3

- 5 a. Explain classfull addressing in detail. (06 Marks)
- b. Explain with neat diagram, the various services provided by network layer. (10 Marks)
- c. Explain datagram approach, with connectionless service. (04 Marks)

OR

- 6 a. Explain datagram format with neat diagram. (10 Marks)
- b. Explain the operation of DHCP with neat diagram, also draw the FSM for the DHCP client. (10 Marks)

Module-4

- 7 a. Explain stop and wait protocol in flow diagram with neat diagram. (10 Marks)
b. Explain connectionless and connection oriented protocols in transport layer. (10 Marks)

OR

- 8 a. Explain Go-back-N protocol, along with sliding window diagrams. (10 Marks)
b. Explain Time-line diagram for a common scenario. (10 Marks)

Module-5

- 9 a. With neat diagram, explain the logical connection at the application layer. (10 Marks)
b. Explain the formats of the request and response message. (06 Marks)
c. Explain FTP with a neat diagram. (04 Marks)

OR

- 10 a. Explain about electronic-mail architecture. (10 Marks)
b. Explain DNS Resolution and its types: (i) recursive resolution and (ii) iteration resolution. (10 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2023/Jan.2024 Electromagnetic Waves

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State and explain coulomb's law of force between two point charges in vector form. (06 Marks)
- b. Convert point P(1, 3, 5) to cylindrical and spherical co-ordinates. Also write the equations for differential surface, differential volume for rectangular, cylindrical and spherical systems. (06 Marks)
- c. Find electric field intensity at P(1, 1, 1) caused by 4 identical 3nc charges are located at P₁(1, 1, 0), P₂(-1, 1, 0), P₃(-1, -1, 0) and P₄(1, -1, 0). (08 Marks)

OR

- 2 a. Define electric field intensity. Derive an expression for electric field intensity due to infinite line charge. (08 Marks)
- b. A point charge of 50nc each are located at A(1, 0, 0), B(-1, 0, 0), C(0, 1, 0) and D(0, -1, 0) in free space. Find the total force on the charge at A. Also find $\vec{E}$ at A. (06 Marks)
- c. A uniform line charge $\rho_L = 25nc/m$ lies on the line $x = -3m$, $y = 4m$ in freespace. Find electric field intensity at a point (2, 3, 15)m. (06 Marks)

Module-2

- 3 a. State and prove Gauss's law. (06 Marks)
- b. Evaluate both sides of the divergence theorem for the defined plane in which $1 \leq x \leq 2$, $2 \leq y \leq 3$, $3 \leq z \leq 4$, if $\vec{D} = 4x \hat{a}_x + 3y^2 \hat{a}_y + 2z^3 \hat{a}_z$ c/m². (10 Marks)
- c. Derive the point form of continuity of current equation. (04 Marks)

OR

- 4 a. Obtain the expression for the work done in moving a point charge in an electric field. (06 Marks)
- b. Given that the field $\vec{D} = \frac{5 \sin \theta \cos \phi}{r} \hat{a}_r$ c/m². Find : i) Volume charge density ii) The total electric flux leaving the surface of the spherical volume of radius 2m. (08 Marks)
- c. Define potential difference. Derive the expression for potential field of a point charge. (06 Marks)

Module-3

- 5 a. State and prove uniqueness theorem. (08 Marks)
- b. Define Stoke's theorem. Use this theorem to evaluate both sides of the theorem for the field $\vec{H} = 6xy \hat{a}_x - 3y^2 \hat{a}_y$ A/m and the rectangular path around the region, $2 \leq x \leq 5$, $-1 \leq y \leq 1$ and $z = 0$. Let the positive direction of ds be $\hat{a}_z$. (12 Marks)

OR

- 6 a. Solve the Laplace's equation for the potential field in the homogeneous region between the two concentric conducting spheres with radii 'a' and 'b' such that $b > a$, if potential $v = 0$ at $r = b$ and $v = v_0$ at $r = a$. Also find the capacitance between concentric spheres. (08 Marks)
- b. Derive the expression for magnetic field intensity due to infinite long straight conductor using Biot-Savart's law. (06 Marks)
- c. Determine whether or not the following potential fields satisfy the Laplace's equation:
 i) $V = 2x^2 - 3y^2 + z^2$ ii) $V = r \cos\theta + \phi$ (06 Marks)

Module-4

- 7 a. Derive an expression for Lorentz Force equation. (06 Marks)
- b. If $\vec{B} = 0.05x \hat{a}_y$ Tesla in a material for which $\mu_m = 2.5$, Find: i) μ_r ii) μ iii) $\vec{H}$ iv) $\vec{M}$
 v) $\vec{J}$ vi) $\vec{J}_b$. (08 Marks)
- c. Derive the expression for the force between two differential current elements. (06 Marks)

OR

- 8 a. Derive the expression for the boundary conditions between two magnetic medias. (10 Marks)
- b. Calculate the magnetization in magnetic material where:
 i) $\mu = 1.8 \times 10^5$ H/m and $M = 120$ A/m
 ii) $\mu_r = 22$, there are 8.3×10^{28} Atoms/m³ and each atom has a dipole moment of 4.5×10^{-27} A/m²
 iii) $B = 300$ μ T and $\chi_m = 15$. (06 Marks)
- c. Briefly explain the forces on magnetic materials. (04 Marks)

Module-5

- 9 a. List and explain Maxwell's equations in point form and integral form. (08 Marks)
- b. Given $\vec{E} = E_m \sin(\omega t - \beta z) \hat{a}_y$ V/m. Find: i) $\vec{D}$ ii) $\vec{B}$ iii) $\vec{H}$. Sketch $\vec{E}$ and $\vec{H}$ at $t = 0$. (08 Marks)
- c. Find the frequency at which conduction current density and displacement current density are equal in a medium with $\sigma = 2 \times 10^{-4}$ mho/m and $\epsilon_r = 81$. (04 Marks)

OR

- 10 a. State and prove Poynting theorem. (08 Marks)
- b. For the given medium $\epsilon = 4 \times 10^{-9}$ F/m and $\sigma = 0$, find 'K' so that $\vec{E} = (20y - kt) \hat{a}_x$ V/m and $\vec{H} = (y + 2 \times 10^6 t) \hat{a}_z$ A/m. (06 Marks)
- c. A uniform plane wave of frequency 10MHz travels in positive direction in a lossy medium with $\epsilon_r = 2.5$, $\mu_r = 4$ and $\sigma = 10^{-3}$ Ω /m. Calculate α , β , γ and η , λ . (06 Marks)

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18EC71

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the functions of each layer in TCP/IP protocol suite. (10 Marks)
- b. Explain the components of a data communication. (05 Marks)
- c. Illustrate the concept of multiplexing and demultiplexing at the upper three layers of TCP/IP protocol suite. (05 Marks)

OR

- 2 a. Explain four physical topologies of a network. (10 Marks)
- b. With a neat diagram illustrate the concepts of encapsulation and decapsulation in internet. (10 Marks)

Module-2

- 3 a. Describe the operation of stop and wait protocol with FSM and flow diagram. (12 Marks)
- b. Define ARP and its position in TCP/IP protocol suite and also explain ARP operation with relevant diagram. (08 Marks)

OR

- 4 a. Explain how collisions are avoided through the use of CSMA/CA's three strategies with flow diagram. (10 Marks)
- b. Explain briefly 10 Base 5 and 10 Base T implementation. (06 Marks)
- c. A slotted ALOHA network transmits 200 bit frames using a shared channel with a 200 kbps bandwidth. Find the throughput if the system (all the stations together) produces.
 - (i) 1000 frames per second
 - (ii) 500 frames per second
 - (iii) 250 frames per second(04 Marks)

Module-3

- 5 a. Compare and contrast connectionless packet-switched network with a virtual-circuit packet switched network using necessary diagrams. (08 Marks)
- b. An organization is granted a block of addresses with beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets, one sub-block of 10 addresses, one subblock of 60 addresses and one subblock of 120 addresses. Design the subblocks. (06 Marks)
- c. Explain MPLS packet, briefly. (06 Marks)

OR

- 6 a. Illustrate IPv4 datagram format. (10 Marks)
- b. Explain path-vector routing by using spanning tree. Also apply path-vector algorithm for updating path-vectors. (10 Marks)

Module-4

- 7 a. Explain FSMs for Go-Back-N protocol with a neat diagram. (08 Marks)
b. Explain the concept of sliding window in circular and linear formats with suitable figures. (07 Marks)
c. Explain why the size of the Sender and Receiver windows chosen as one half of 2^m for Selective Repeat Protocol. (05 Marks)

OR

- 8 a. Explain TCP segment format. (08 Marks)
b. Illustrate connection establishment in TCP using Three-way handshaking using suitable example. (07 Marks)
c. Explain briefly Tahoe TCP with FSM. (05 Marks)

Module-5

- 9 a. Explain non-persistent connection with suitable example. (08 Marks)
b. Explain client-server paradigm with an example. (06 Marks)
c. Describe the three Mail Transfer Phases. (06 Marks)

OR

- 10 a. Explain DNS resolution and its type Recursive Resolution. (07 Marks)
b. Explain briefly local versus remote lagging in Telnet with a neat diagram. (07 Marks)
c. Describe the components of SSH. (06 Marks)

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Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 VLSI Design

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Missing data may be suitably assumed.

Module-1

- 1 a. Derive an expression for drain current in linear and saturation region. (08 Marks)
- b. Draw the CMOS inverter circuit and explain its D.C. characteristic. (08 Marks)
- c. Implement a 2:1 MUX using transmission gate. (04 Marks)

OR

- 2 a. Explain the non ideal IV effect of MOSFET with respect to CMOS channel length modulation and mobility degradation. (08 Marks)
- b. Explain the operation of nMOS transistor with IV characteristics. (08 Marks)
- c. Sketch a static CMOS gate computing $y = \overline{(A + B + C)D}$. (04 Marks)

Module-2

- 3 a. Explain CMOS nWell process with necessary diagrams. (12 Marks)
- b. Mention different types of MOSFET capacitances with necessary diagrams and equations also MOSFET. Capacitances in cut off, linear and saturation region. (08 Marks)

OR

- 4 a. Define scaling. Explain constant field scaling and constant voltage scaling and why constant voltage scaling is usually preferred over full scaling. (07 Marks)
- b. With neat diagram, explain the Lambda based design rules for two metal layers. (06 Marks)
- c. Draw the layout for $f = \overline{ABC}$ and estimate the cell area. (07 Marks)

Module-3

- 5 a. Develop the RC delay model to compute the delay of the logic circuit and calculate the delay of unit sized inverter driving another unit in vertex. (06 Marks)
- b. Estimate t_{pdf} and t_{pdr} for the 3 input NAND gate shown in Fig.Q.5(b) if the output is loaded with h identical NAND gates. (08 Marks)

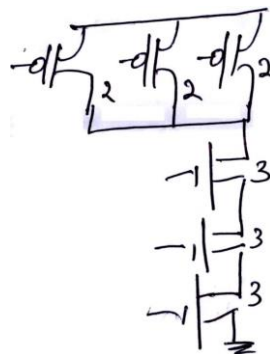


Fig.Q.5(b)

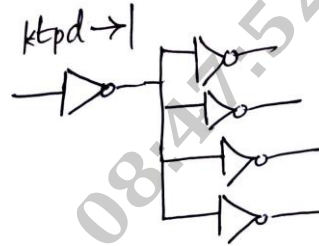
- c. Explain eVSL with an example.

(06 Marks)

OR

- 6 a. Explain: i) Pseudo-nMOS ii) Ganged CMOS with necessary circuit examples. (06 Marks)
- b. If a unit transistor has $R = 10K\Omega$ and $c = 0.1fF$ in a 65nm process, compute the delay, in picoseconds, of the inverter Fig.Q.6(b) with a fan out of $h = 4$. (06 Marks)

Fig.Q.6(b)

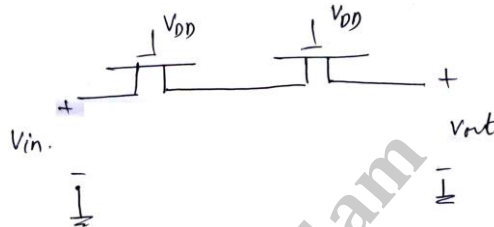


- c. Explain linear delay model compare the logical effort of the following gates with the help of schematic diagrams: i) 3-input NAND gate ii) 3 input NOR gate. (08 Marks)

Module-4

- 7 a. Explain Resettable latches and flipflops using CMOS transmission gate. (06 Marks)
- b. Explain Dynamic logic. (06 Marks)
- c. Consider the two nFET chain in Fig.Q.7(c). The power supply is set to a value of $V_{DD} = 3.3V$ and the nFET threshold voltage is $V_{Tn} = 0.55V$. Find the output voltage V_{out} at the right side of the chain for the following values: i) $V_{in} = 2.9V$ ii) $V_{in} = 3.0V$ iii) $V_{in} = 1.4V$ iv) $V_{in} = 3.1V$. (08 Marks)

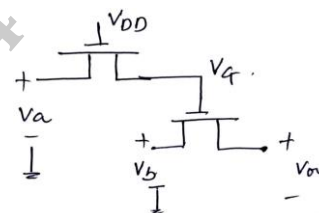
Fig.Q.7(c)



OR

- 8 a. Explain pulsed latches with schematic and waveforms. (06 Marks)
- b. The output of an nFET is used to drive the gate of another nFET as shown in Fig.Q.8(b). Assume that $V_{DD} = 3.3V$ and $V_{tn} = 0.6V$. Find the output voltage V_{out} when the input voltages are at following values:
i) $V_a = 3.3V$ and $V_b = 3.3V$
ii) $V_a = 2.0V$ and $V_b = 2.5V$. (08 Marks)

Fig.Q.8(b)



- c. Explain Domino logic. (06 Marks)

Module-5

- 9 a. With neat schematic diagram explain the operation of Full CMOS static RAM cell. (10 Marks)
- b. Explain the different fault models. (10 Marks)

OR

- 10 a. With neat schematic diagram explain the operation of three transistor DRAM cell. (10 Marks)
- b. Write short notes on: i) Built in Self Test ii) Scan Design. (10 Marks)

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18EC733

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024

Digital Image Processing

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the working of sensor strips and discuss the applications in airborne imaging and medical imaging with neat sketches. (08 Marks)
- b. Define 4, 8 and m connectivity. Compute the lengths of shortest 4, 8 and m paths between the pixels p and q in the image segment shown in Fig.Q1(b) by considering intensity set $v = \{2, 3, 4\}$.

3	4	1	2
0	1	4	2 (q)
2	2	3	4
3 (p)	0	4	2

Fig.Q1(b)

(06 Marks)

- c. Explain the components of an image processing system with a neat block diagram. (06 Marks)

OR

- 2 a. Describe Ultra Sound (US) imaging with any one example (medical/industry). Also explain the methods of image formation used in US imaging. (10 Marks)
- b. Demonstrate with experiments, how perceived image quality varies with spatial and gray level resolutions and discuss your observations with a neat graph on NK plane (Isopreference curve). (10 Marks)

Module-2

- 3 a. Explain the following gray level transformations :
 - i) Gray level slicing
 - ii) Bit plane slicing. (08 Marks)
- b. What is meant by histogram matching? Develop a probabilistic model for continuous and discrete functions to demonstrate histogram matching. (10 Marks)
- c. Discuss local histogram processing. (02 Marks)

OR

- 4 a. Explain image sharpening in the spatial domain using second order derivative filter. (Use Laplacian operator). (08 Marks)
- b. Determine histogram matched values for the given input image and target histogram as shown in Table Q4(b).

r_i	n_i	$P_z(z_q)$
0	790	0.0
1	1023	0.0
2	850	0.0
3	656	0.15
4	329	0.2
5	245	0.3
6	122	0.2
7	81	0.15

Table Q4(b)

Here $r_i \rightarrow i^{\text{th}}$ intensity of input image

$n_i \rightarrow$ number of pixels i^{th} having intensity level.

$P_z(z_e) \rightarrow$ Target histogram

Given $n \rightarrow$ total number of pixels in an input image is 4096.

(12 Marks)

Module-3

- 5 a. Define 2D – DFT of an image $f(x, y)$ and its inverse DFT. Also state the following properties of 2D – DFT.
- Translation
 - Rotation
 - Periodicity
 - 2D convolution.
- (08 Marks)
- b. Describe smoothing frequency domain filters, for image enhancement. Also explain the working of following filters for image smoothing in frequency domain :
- Ideal LPF
 - Butterworth LPF
 - Gaussian LPF.
- (08 Marks)
- c. Explain selective filtering using band reject filters. (04 Marks)

OR

- 6 a. Explain the basic procedure used for filtering in frequency domain. (06 Marks)
- b. Explain the working of homomorphic filtering in image processing using mathematical equations and response. (08 Marks)
- c. State and prove the conjugate symmetry properties of 2D – DFT with respect to an image $f(x, y)$. (06 Marks)

Module-4

- 7 a. Explain the module of the image degradation/restoration process. (06 Marks)
- b. Describe how the images are restored in the presence of only noise interference. Also explain the following mean filters used for image restoration.
- Arithmetic mean
 - Geometric mean
 - Harmonic mean
 - Contra harmonic mean.
- (10 Marks)
- c. Explain inverse filtering with necessary mathematical equations and examples. (04 Marks)

OR

- 8 a. Explain the following noise Probability Density Functions (PDF) used in image processing.
- i) Gaussian
 - ii) Rayleigh
 - iii) Gamma
 - iv) Exponential
 - v) Uniform
 - vi) Impulse.
- (12 Marks)
- b. What are adoptive filters? Explain adoptive local noise reduction and adoptive median filter with the algorithms.
- (08 Marks)

Module-5

- 9 a. With a neat sketch, explain color chromaticity diagram. (08 Marks)
- b. Describe the process of RGB to HSI conversions with mathematical equations. (06 Marks)
- c. What is meant by mathematical morphology? Explain dilation and erosion operations using mathematical equations. (06 Marks)

OR

- 10 a. Discuss the process of converting HSI to RGB with relevant mathematical expressions. (10 Marks)
- b. Demonstrate the working operating and closing morphological operations using mathematical equations and real time examples. (08 Marks)
- c. Write a brief note on Pseudo color image processing. (02 Marks)

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18EC741

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 IOT and Wireless Sensor Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List the entities behind the IOT technology and give examples in each case. (10 Marks)
- b. Represent the following:
 - i) Direct and indirect accesses between CoAP client and CoAP server.
 - ii) CoAP client access for resource using resource directory.
 - iii) CoAP client and server access using proxies. (10 Marks)

OR

- 2 a. Represent the following:
 - i) CoAP request or response communication to a machine, IoT device or MT.
 - ii) A computer or machine interface using IP communication to a mobile service provider.
 - iii) A machine or IOT device or MO communication of CoAP request or response communication. (10 Marks)
- b. Write the equations that represent the simple conceptual framework of IOT and complex conceptual framework for IOT using cloud platform based processes and services. (10 Marks)

Module-2

- 3 a. Illustrate the different classes of internet network and for what reason subnet masks are used? (10 Marks)
- b. Describe the different classification of cloud service models and give the example in each case. (10 Marks)

OR

- 4 a. Show the details of IP packets received or transmitted at or to network layer. (10 Marks)
- b. Name two open source platform used in the IOT based data collection, storage and computing. Give the reasons for using these platform. (10 Marks)

Module-3

- 5 a. Write the program for Ardecino controlled traffic light control system in which green light is ON towards East-West in the starting. Three traffic lights-Red, Yellow and Green needs to be controlled on each of the north, east, south and west in clockwise pathways. Assume delays 5s each between successive states of LEDS and steady state for 70s for a pair of pathways. (12 Marks)
- b. Illustrate the layered attacker model and possible attacks using IETF six layer modified model for IOT/M2M. (08 Marks)

OR

- 6 a. Write the programming of Arduino for usages of RFID serial data reading using UART port. Use port 2 and 5 for serial RX and serial TX. (12 Marks)
b. Mention the five levels of software development for application and services for IOT/M2M (no need to draw the block diagram and explanation). (08 Marks)

Module-4

- 7 a. Explain any five required mechanisms in case of wireless sensor network. (10 Marks)
b. Illustrate the energy consumption in microcontrollers used in WSN. (10 Marks)

OR

- 8 a. State the difference between microcontrollers versus microprocessor, FPGAS and ASICS. (10 Marks)
b. What are the 3 types of mobility in WSN scenario and illustrate. (10 Marks)

Module-5

- 9 a. Compare different modulation schemes available in the design of physical layer and transceiver in WSNs. (10 Marks)
b. Explain Perkins model of assignment of MAC address in WSN. (10 Marks)

OR

- 10 a. Explain the general concept of geographic routing. (10 Marks)
b. Illustrate the S-MAC principle used in the low duty cycle protocols and wakeup concept. (10 Marks)

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18EC743

Seventh Semester B.E. Degree Examination, Dec.2023/Jan.2024 Multimedia Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Multimedia. Explain:
(i) Telephone Network
(ii) Integrated Services Digital Network, in detail with neat figure. (10 Marks)
b. Briefly explain the network QoS associated with circuit switched and packet switched networks. (06 Marks)
c. Derive the maximum block size that should be used over a channel which has mean BER probability of 10^{-4} if the probability of a block containing error and hence being discarded is to be 10^{-1} . (04 Marks)

OR

- 2 a. Explain Simplex, Duplex, Broadcast and Multicast operational modes of communication with diagram. (10 Marks)
b. Explain the working principle of circuit mode and packet mode of operation in multimedia networks. List the salient features of each type. (10 Marks)

Module-2

- 3 a. With a neat block diagram, explain PCM signal encoding and decoding method. (10 Marks)
b. Discuss the different types of text representation. (10 Marks)

OR

- 4 a. With neat diagrams, describe the following digitization formats:
(i) 4 : 2 : 2 (ii) QCIF (10 Marks)
b. Explain the following:
(i) Aspect ratio (ii) Quantization Intervals (04 Marks)
c. Derive the time to transmit the following digitized images at both 64 Kbps and 1.5 Mbps.
(i) A $640 \times 480 \times 8$ VGA compatible images.
(ii) A $1024 \times 768 \times 24$ SVGA compatible images (06 Marks)

Module-3

- 5 a. Explain in detail each stage of JPEG encoder with neat diagram. (10 Marks)
b. Describe the function of multimedia operating system with respect to real time processing and QoS based resource management. (10 Marks)

OR

- 6 a. Explain Huffman coding procedure for encoding to the given data "AAAABBCD". (10 Marks)
b. Define distributed multimedia system with neat block diagram, also highlight the salient features. (10 Marks)

Module-4

- 7 a. Explain H.261 video encoder principles with neat diagram. (10 Marks)
- b. Explain the following terms:
- (i) Motion estimation
 - (ii) Motion compensation
 - (iii) I frame, B and D frames
 - (iv) GOP
 - (v) Prediction span
- (10 Marks)

OR

- 8 a. Explain how better sound quality can be obtained using subband coding ADPCM with the help of block diagram and signal encoder and decoder. (10 Marks)
- b. A digitized video is to be compressed using MPEG-1 standard. Assuming a frame sequence of IBBPBBPBBPBBI... and average compression ratio of 10:1(I), 20:1(P) and 50:1(B), derive average bit rate that is generated by encoder for both NTSC and PAL formats. (10 Marks)

Module-5

- 9 a. With an example, explain fragmentation and reassembly in internet. (10 Marks)
- b. Explain datagram format of IPV6 with extension headers. (10 Marks)

OR

- 10 a. Discuss ARP/RARP message formats and transmission. (10 Marks)
- b. Explain distance vector routing protocol with an example. (10 Marks)

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